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Assignment 5 solution

Q1. (a) $A=\begin{pmatrix}1 & 1\\ 1+\delta & 1\end{pmatrix}$, $\delta>0$

$\|A\| = 2+\delta$

$A^{-1} = -\frac{1}{\delta}\begin{pmatrix}1 & -1\\ -1-\delta & 1\end{pmatrix}$

$\|A^{-1}\| = \frac{2+\delta}{\delta}$

$\therefore \text{cond}(A) = \|A\|\cdot\|A^{-1}\| = \frac{(2+\delta)^2}{\delta}$

(b) $r = b - Ax_a = \begin{pmatrix}2\\ 2+\delta\end{pmatrix} - \begin{pmatrix}1 & 1\\ 1+\delta & 1\end{pmatrix}\begin{pmatrix}-1\\ 3+\delta\end{pmatrix} = \begin{pmatrix}-\delta\\ \delta\end{pmatrix}$

relative backward error:

$$\frac{\|r\|_\infty}{\|b\|_\infty} = \frac{\delta}{2+\delta}$$

relative forward error:

$x=\begin{pmatrix}1\\ 1\end{pmatrix}$ ↗ the solution of $Ax=b$

$$\frac{\|x-x_a\|_\infty}{\|x\|_\infty} = \frac{\left\|\begin{pmatrix}2\\ -2-\delta\end{pmatrix}\right\|_\infty}{1} = 2+\delta$$

$\therefore$ error magnification $= \frac{2+\delta}{\frac{\delta}{2+\delta}}$

$= \frac{(2+\delta)^2}{\delta}$

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Q2. Solution:

Rearrange: $4u + 3w = 0$

$u + 4v = 5$

$v + 2w = 2$

$$\begin{pmatrix} 4 & 0 & 3 \\ 1 & 4 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ 2 \end{pmatrix}$$

(i) Jacobi, initial guess $[0, 0, 0]$

$u = -\frac{3}{4}w$

$v = \frac{1}{4}(5 - u)$

$w = \frac{1}{2}(2 - v)$

So $x^{(1)} = \begin{pmatrix} -\frac{3}{4} \cdot 0 \\ \frac{1}{4}(5-0) \\ \frac{1}{2}(2-0) \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{5}{4} \\ 1 \end{pmatrix} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$

$x^{(2)} = \begin{pmatrix} -\frac{3}{4} \cdot 1 \\ \frac{1}{4}(5-0) \\ \frac{1}{2}(2-\frac{5}{4}) \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} \\ \frac{5}{4} \\ \frac{3}{8} \end{pmatrix}$

(ii) Gauss-Seidel, initial guess $[0, 0, 0]$

$u = -\frac{3}{4}w$

$v = \frac{1}{4}(5 - u)$

$w = \frac{1}{2}(2 - v)$

So $\begin{pmatrix} u^{(1)} \\ v^{(1)} \\ w^{(1)} \end{pmatrix} = \begin{pmatrix} -\frac{3}{4}w^{(0)} \\ \frac{1}{4}(5 - u^{(1)}) \\ \frac{1}{2}(2 - v^{(1)}) \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{5}{4} \\ \frac{3}{8} \end{pmatrix}$

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$$\text{so} \begin{pmatrix} u^{(2)} \\ v^{(2)} \\ w^{(2)} \end{pmatrix} = \begin{pmatrix} -\frac{3}{4}w^{(1)} \\ \frac{1}{4}(5-u^{(1)}) \\ \frac{1}{2}(2-v^{(1)}) \end{pmatrix} = \begin{pmatrix} -\frac{9}{32} \\ \frac{169}{128} \\ \frac{87}{256} \end{pmatrix}$$

Q3. Proof:

(a) $$x_i^{(k+1)} = \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^{n} a_{ij} x_j^{(k)}\right) / a_{ii}$$

$$= \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij} x_j^{(k)} + a_{ii} x_i^{(k)}\right) / a_{ii}$$

$$= r_i^{(k+1)} + x_i^{(k)}$$

(b) $$\varepsilon_i^{(k+1)} - \varepsilon_i^{(k)} = x_i^{(k+1)} - x_i^* - (x_i^{(k)} - x_i^*)$$

$$= x_i^{(k+1)} - x_i^{(k)}$$

$$= r_i^{(k+1)} / a_{ii} \quad (\text{由(a)})$$

$$= \left(b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij} x_j^{(k)}\right) / a_{ii}$$

$$= \left(\sum_{j=1}^{n} a_{ij} x_j^* - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i}^{n} a_{ij} x_j^{(k)}\right) / a_{ii}$$

$$= \left(\sum_{j=1}^{i-1} a_{ij} (-\varepsilon_j^{(k+1)}) + \sum_{j=i}^{n} a_{ij} (-\varepsilon_j^{(k)})\right) / a_{ii}$$

$$= -\frac{R_i^{(k+1)}}{a_{ii}}$$